STACKS, DISCRIMINANTS: LEMMA-LOCALIZE-DUALIZING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 379-388 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 390-397, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 46 of queue_discriminant.json (pb=415, stmt 379-388, proof 390-397).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0583: sources/stacks-discriminant/pinned-original-source.tex, source lines 397–397. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $A \to B$ be a quasi-finite map of Noetherian rings.*

(1) *If $A \to B$ factors as $A \to A_f \to B$ for some $f \in A$, then $\omega_{B/A} = \omega_{B/A_f}$.*
(2) *If $g \in B$, then $(\omega_{B/A})_g = \omega_{B_g/A}$.*
(3) *If $f \in A$, then $\omega_{B_f/A_f} = (\omega_{B/A})_f$.*

Proof. Say $A \to B' \to B$ is a factorization with $A \to B'$ finite and $\mathrm{Spec}(B) \to \mathrm{Spec}(B')$ an open immersion. In case (1) we may use the factorization $A_f \to B'_f \to B$ to compute ω_{B/A_f} and use Algebra, Lemma 0583. In case (2) use the factorization $A \to B' \to B_g$ to see the result. Part (3) follows from a combination of (1) and (2). □

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.